

Life on Silicon: Toward Autonomous Synthetic Agents

David Warutumo (davidwarchy@gmail.com)

Abstract

The *Life on Silicon* vision seeks to create autonomous synthetic agents capable of perceiving, moving, manipulating, and evolving within the physical world. These agents will not rely on human supervision or dense computation, but instead learn fast, heuristic strategies for survival and goal achievement. This initiative sits at the intersection of robotics, artificial life, and embodied intelligence - aiming ultimately to engineer adaptive, evolving lifeforms in silicon rather than carbon.

1. Introduction

Autonomy emerges when an agent can sense, act, and adapt to its environment in pursuit of its own persistence. The *Life on Silicon* project frames this as an engineering and philosophical challenge: to construct systems that exhibit lifelike adaptation and survival dynamics under real-world constraints.



2. Core Dimensions of Autonomy

1. **Perception** – Development of minimal yet sufficient sensory systems for meaningful environmental interaction.
 2. **Motion** – Mechanisms by which agents move within the world, linking perception to action and enabling goal pursuit.
 3. **Manipulation** – Handling and transforming objects, particularly those vital to the agent's own operation or continuity.
 4. **Activity** – Structuring sequences of manipulations into coherent, productive routines that yield adaptive outcomes.
 5. **Selection** – Determining how agents evolve or are shaped under physical and environmental pressures. Should intelligence emerge through centralized design or decentralized evolution?
-

3. Evolutionary Axes

- **Morphology** – Dynamic body structures are key to enduring intelligence. Changing morphologies enable new forms of sensing, movement, and interaction—driving functional evolution.
 - **Control** – Diverse control strategies promote robustness. The goal is to discover practical algorithms that can autonomously generate and sustain this diversity in behavior and decision-making.
-

4. Current Focus: Motion and Navigation

The immediate research focus lies in **motion**, particularly **navigation**. We aim to develop agents that:

- Move toward beneficial stimuli (object-goal navigation).
- Avoid harmful stimuli (defensive or evasive motion).

This requires an implicit understanding of spatial relationships—but not necessarily detailed metric reconstruction. Unlike high-precision SLAM systems such as [MASt3R-SLAM](#), this research explores whether agents can learn **rapid, heuristic navigation strategies** that are computationally lightweight yet behaviorally effective.

Illustrative Scenario:

A factory agent carries fragile loads across a cluttered warehouse with minimal visual or positional cues. If the floor becomes uneven and obstacles move unpredictably, the agent must adapt in real time - relying not on maps, but on emergent heuristics shaped by experience.

This research thus occupies a unique space between **robotics**, **object-goal navigation**, and **active SLAM**, targeting the minimal conditions for spatial competence and adaptive motion in embodied agents.

5. Broader Vision

By integrating perception, motion, manipulation, activity, and selection—while enabling the evolution of morphology and control—the *Life on Silicon* initiative outlines a complete framework for autonomy. The end goal is the creation of **synthetic life**: agents that learn, adapt, and evolve, embodying intelligence and persistence not as software, but as *living systems in silicon*.